

Generation Interconnection System Impact Study Report

For Queue Project Q21-1794

Rush Creek 138 kV

July 24, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1794. The Interconnection Customer proposes a 159.4 MW solar photovoltaic facility interconnecting at the Rush Creek 138 kV substation in the Virginia area. The request was received on June 24, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 409 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,444,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Pahokee Substation – Zephyrhills 138 kV L is overloaded to 102.2% of its 231 MVA emergency rating for the loss of the Easton 2 – Cady 345 kV. The project contributes 6.1% DFAX (9.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pahokee Substation – Zephyrhills 138 kV L	Reconductor line	\$1,444,000	100% (\$1,444,000)
	Total allocated to project		\$1,444,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.